

Module 4: Relevant Costing

Problems:

2) Calculation of relevant cost of materials for the special order

Material	Nature & Computation	Amount (₹)
A	1000 units to be purchased. Here, out of pocket cost is relevant. The relevant cost (1000 units $\times$ ₹ 6 per unit)	6000
B	1000 units is to be replaced. Here, replacement cost is relevant. The relevant cost (1000 units $\times$ ₹ 5 per unit)	5000
C	700 units at Net Realizable value, here opportunity cost is relevant i.e., 700×2.5 300 units to be purchased, here out of pocket cost is relevant (300×4)	1750 1200
D	200 units at Net Realizable value, i.e., $200 \text{ units} \times 6 = 1200$ Substitute for material E $300 \text{ units} \times 5 = 1500$ here, highest of the opportunity cost is relevant	1200 1500
	total relevant cost of materials	15,450

The order can be accepted since the relevant cost of material is lesser than the price of the special order provided the labour cost & overheads remain the same.

3) Computation of Relevant cost of Machine	
Nature and Computation	Amnt(₹)
variable cost of operating the machine is relevant	60000
Reduction in NRV as opportunity cost is relevant (40000 - 25000)	15000
Total relevant cost	<u>75000</u>

4) Computation of Relevant cost of the Special cost		
Items	Nature & computation	Amnt(₹)
Equipment	Reduction in NRV as opportunity cost is relevant (1500-800)	700
Storage	Additional rent as out of pocket cost is relevant	18000
Sub-Assembly	Cost of making is $(10+6+6)=22$ and cost of buying is ₹ 24. ∴ It is better to make the assembly	22 p.u

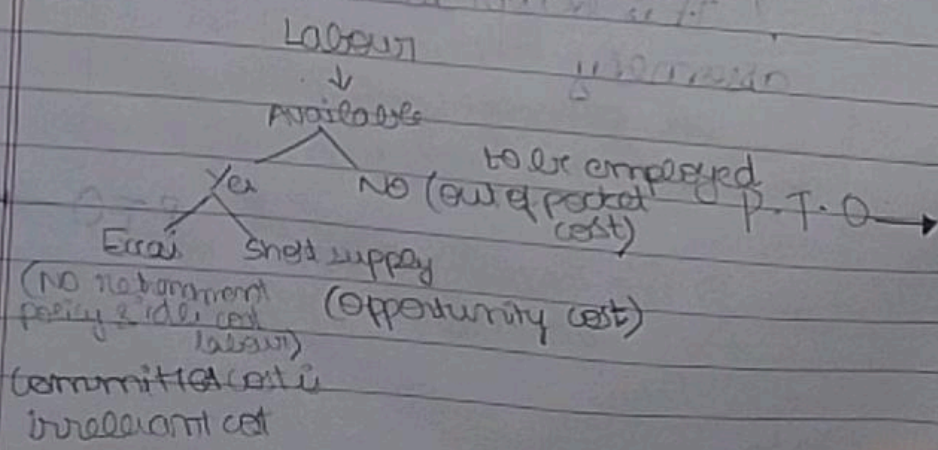
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5) Computation of opportunity cost

Particulars	X	Y	Z
Selling price	10	12	12
less: variable cost	6	9	7
Contribution per unit	4	3	5
Market Demand (units)	3000	2000	1000
Production capacity (units)	2000	3000	9000
Effective production (whichever is lower)	2000	2000	900
Total contribution	8000	6000	4500
Opportunity cost	6000	8000	8000

Particulars	Alternative I Make	Alternative II Buy & own with plant making and out the machine	Alternative III Buy and hire out the machine
Variable cost	10000 units x Rs. 18 p.u. 180000	10000 units x Rs. 19 190000	10000 units x Rs. 19 p.u. 190000
Relevant benefit	-	10000 units x Rs. 2 p.u. 20000	-
Net Relevant cost	180000	170000	169000

Conclusion: Third alternative is the best option as it gives the lowest Net Relevant cost.



1 → 1 hr
20 per hour

10 × 180 = 18000
→ 20000
Rent → 21000

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Particulars	Nature and Computation	Amount (₹)
Material M	Material M is a stock-making item. Therefore, NRV as opportunity cost is relevant.	₹. 80
Labour	Permanent employee and the redeployment does not result in loss of contribution. Hence, here the committed cost is irrelevant.	Nil
Out-of-pocket expenses	It is a relevant cost.	₹. 30
	Minimum price to be charged to the customer	110

8)

Particulars	Nature & Computation
Relevant costs	Operating cost of old machine ₹. 100 per hour Cost of the new machine - ₹ 3500000 Operating cost of the new machine - ₹ 70 per hr NRV of old machine - ₹ 1200000 Revenue from both the machines - ₹ 120 per hour

No. of hrs the new machine has to run for the benefit of the company

Relevant Benefit =

$$35L - 12L = 23L$$

$$\text{Contribution per hr} = 120 - 70 = ₹ 50 \text{ per hr}$$

$$23000000 / 50 \text{ per hr} = 460000 \text{ hrs}$$

BEP of both
the machines

$$\text{BEP Machine 1} = 20000000 / 120 - 100$$

$$= 100000 \text{ hrs}$$

$$\text{BEP machine 2} = 35000000 / 120 - 70$$

$$= 70000 \text{ hrs}$$

12)

Particulars
Relevant cost

Nature & computation

Operating cost of the old machine =
£ 130 per hour

Cost of the new machine = 4500000

Operating cost of the new machine
= £ 91 per hour

Revenue from machine per hour
= £ 195

NRV of old machine = £ 1560000

Relevant Benefit = 4550000 - 1500000
= 2990000

No. of hrs the new
machine has to run
to the benefit of the
company

Contribution per hour = 195 - 91
= 104 per hour

$$2990000 / 104 = 28750 \text{ hrs}$$

BEP of both the
machines

$$\text{BEP machine 1} = 2600000 / 195 - 130$$

$$= 40000 \text{ hrs}$$

$$\text{BEP machine 2} = 4550000 / 195 - 91$$

$$= 43750 \text{ hrs}$$

9. Particulars	Nature and computation
i) Relevant cost	Sale value of shirt in the local market $\text{₹} 120$ per shirt Cost of rectification $\text{₹} 30$ per shirt Sale value after rectification $\text{₹} 160$ per shirt
ii) Should the rectification be done?	Increase in sale value by $\text{₹} 40 (160 - 120)$ Cost of rectification $\text{₹} 30$ Profit of $\text{₹} 10$ per shirt Therefore, rectification should be done.
iii) What is the gain or loss if not rectified?	Loss of $\text{₹} 10$ per shirt $(40 - 30)$ Therefore if not rectified there is a loss of $\text{₹} 200000 (10 \times 20000)$ ^{shirts}
iv) Gain or loss if rectified	Gain of $\text{₹} 10$ per shirt $= 200000 \text{ ₹}$

11. Particulars	Nature and computation
i) Relevant cost	Sale value of the shirt in the local market $\text{₹} 240$ per shirt Cost of rectification is $\text{₹} 60$ per shirt Sale value after rectification is $\text{₹} 320$ per shirt
ii) Should the rectification be done?	Increase in sale value by $\text{₹} 80 (320 - 240)$ Cost of rectification $\text{₹} 60$ Profit of $\text{₹} 20$ per shirt $\therefore$ the rectification should be done

iii) What is the gain or loss if rectification is not done

Loss of £20 per shirt (80-60)
Therefore, if it is not rectified then there will be a loss of £200000 (20 x 10000)

iv) Gain or loss if rectified

Gain of £20 per shirt
= £200000

10) Particulars
Relevant costs

Nature and Computation
Retail value of the paper in inventory 10 dollars
Current market price of 26 dollars

Relevant cost calculation

Opportunity cost at NRV (100 items x 10) = 1000
Out of pocket cost (150 items x 26) = 3900
Total relevant cost = 1000 + 3900 = 4900

13) Cost of making the component of

Particulars	Amount (£)
Raw material	4
Direct labour	8
Production overhead (16 x 60%)	9.6
Cost of making the component	21.6

Cost of buying the component = 20
The company should buy the component rather than making it

14) Computation of relevant cost of the special order

Particulars	Amnt(₹)
Raw materials	8
Direct labour	4
Production overheads (4x50%)	2
Total relevant cost per unit	14
Selling price per unit	16
Additional contribution per unit	2

Accept the special order as this will generate an additional contribution of ₹4000 (₹2 per unit x 2000 units) as the total profit will increase by ₹.4000

15) Computation of relevant cost for the special order

Particulars	Nature & computation	Amnt(₹)
Raw material A	Opportunity cost is a relevant cost	1750
Raw material B	Out-of-pocket cost is a relevant cost	8000
Direct labour	Labour cost of ₹6000 is a committed cost. Hence irrelevant. Extra labour is a relevant cost as out-of-pocket cost	7000
Supervision	Irrelevant	0
Overheads	Irrelevant	0
New machinery	Net cost of machinery (8000-5000)	3000
	Total relevant cost	19750